

ANGELINA COUNTY AP, TX, USA

WMO: 722446

Lat: 31.236N

Lon: 94.754W

Elev: 88

StdP: 100.27

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93987

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.8	-1.2	-10.1	1.6	2.5	-7.7	2.0	4.7	8.8	14.1	8.2	14.7	1.6	0	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	37.2	24.4	35.8	24.6	34.6	24.6	26.5	32.3	26.1	31.8	25.8	31.3	3.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.5	28.1	24.8	20.1	27.9	24.4	19.5	27.7	82.9	32.0	81.2	31.9	79.7	31.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	7.1	6.2	DB	-6.1	38.5	1.9	1.9	-7.5	39.9	-8.5	40.9	-9.6	42.0	-10.9	43.3	(4)
			WB	-7.0	27.2	1.7	0.6	-8.2	27.6	-9.2	27.9	-10.2	28.3	-11.4	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.8	10.0	12.2	15.8	19.6	23.9	27.2	28.5	28.6	25.7	20.3	14.8	10.7
	DBStd	7.85	5.54	5.54	4.92	4.04	3.10	2.06	1.60	1.98	2.87	4.23	5.03	5.34
	HDD10.0	189	71	37	11	1	0	0	0	0	0	1	14	56
	HDD18.3	996	266	183	108	34	3	0	0	0	1	30	127	244
	CDD10.0	3767	69	97	190	288	431	516	573	577	472	318	158	78
	CDD18.3	1533	6	11	29	72	175	266	314	318	223	89	22	8
	CDH23.3	15377	19	53	191	564	1562	2761	3534	3604	2140	788	136	27
	CDH26.7	7097	0	6	27	132	597	1326	1818	1919	1002	252	18	1
Wind	WSAvg	2.7	3.0	3.2	3.3	3.3	3.0	2.6	2.3	2.2	2.2	2.4	2.7	2.7
Precipitation	PrecAvg	1137	104	87	95	90	115	110	68	72	102	94	106	110
	PrecMax	1771	334	256	212	224	308	327	160	216	279	251	406	256
	PrecMin	673	6	12	6	8	0	10	0	4	19	2	17	16
	PrecStd	269	72	54	48	56	73	80	42	57	55	73	73	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	27.3	29.1	31.5	34.3	37.5	38.0	39.6	37.3	32.6	28.6	26.1
		MCWB	18.2	18.4	19.8	21.9	23.5	24.5	24.6	23.7	23.3	23.0	21.6	20.3
	2%	DB	23.4	24.9	27.3	29.6	32.6	35.4	36.6	37.4	34.9	31.1	26.7	23.7
		MCWB	17.6	18.2	19.3	21.2	23.4	24.3	24.6	24.4	23.7	22.4	20.3	19.5
	5%	DB	21.4	22.9	25.7	28.2	31.4	34.0	35.2	35.9	33.4	29.6	24.9	22.0
		MCWB	17.1	17.5	18.4	20.4	23.3	24.5	24.9	24.8	23.6	21.4	19.2	18.6
	10%	DB	19.6	21.1	23.7	26.8	30.1	32.8	34.0	34.6	32.1	27.8	22.9	20.1
		MCWB	16.1	17.0	17.7	19.6	22.8	24.5	24.9	24.8	23.4	21.0	18.9	17.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.6	20.8	21.8	24.1	25.9	26.9	26.9	27.0	26.3	25.5	23.0	21.6
		MCDB	22.8	24.0	26.0	27.6	31.3	31.9	33.0	33.1	31.4	29.7	26.1	23.3
	2%	WB	19.5	20.0	20.8	23.0	25.1	26.2	26.4	26.5	25.7	24.6	21.9	20.6
		MCDB	21.9	22.9	24.3	26.9	30.3	31.5	32.2	32.5	30.7	28.3	24.6	22.8
	5%	WB	18.4	19.1	20.2	22.1	24.4	25.8	26.0	26.1	25.2	23.7	20.9	19.5
		MCDB	20.7	21.8	23.5	26.0	29.1	30.9	31.7	31.9	30.1	27.2	23.5	21.6
	10%	WB	17.0	17.9	19.2	21.4	23.8	25.4	25.7	25.7	24.7	22.7	19.7	17.9
		MCDB	19.1	20.5	22.6	25.0	28.2	30.5	31.3	31.4	29.3	25.9	22.5	19.8
Mean Daily Temperature Range	5% DB	MDBR	11.8	11.6	12.1	12.2	11.2	10.8	11.2	11.7	11.8	12.6	12.2	11.4
		MCDBR	13.6	13.0	12.9	12.4	12.0	12.4	12.9	13.8	13.5	13.5	13.2	12.5
		MCWBR	8.7	7.4	6.3	5.6	4.4	3.7	3.4	3.6	4.3	5.9	7.6	8.5
	5% WB	MCDBR	10.7	10.4	10.5	10.0	10.3	10.4	11.2	11.6	10.8	10.2	11.1	10.4
		MCWBR	8.0	7.1	6.5	5.6	4.6	3.8	3.7	3.7	4.1	5.4	7.6	8.1
Clear-Sky Solar Irradiance	taub	0.326	0.346	0.364	0.410	0.452	0.465	0.482	0.475	0.435	0.370	0.362	0.335	
	taud	2.527	2.481	2.427	2.306	2.229	2.240	2.225	2.239	2.312	2.476	2.446	2.512	
	Ebn at Noon	904	914	918	883	844	829	813	815	836	876	851	872	
	Edh at Noon	86	99	111	131	142	140	142	138	124	98	93	84	
All-Sky Solar Radiation	RadAvg	2.80	3.12	4.34	5.34	5.85	6.21	6.24	5.91	5.05	4.19	3.09	2.43	
	RadStd	0.26	0.55	0.40	0.34	0.43	0.47	0.47	0.45	0.47	0.57	0.34	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+109	+90

Nomenclature: See separate page